NOTE: The document identifier and heading has been changed on this page to reflect that this is a performance specification. There are no other changes in this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-17695B(SHIPS)
15 October 1968
SUPERSEDING
MIL-I-17695A(SHIPS)
9 November 1960
(See 6.3 and 6.4)

PERFORMANCE SPECIFICATION

INSULATION TAPE, ELECTRICAL, FILLER TYPE,
FLAMEPROOF, SYNTHETIC

1. SCOPE

1.1 This specification covers the requirements for a filler type insulation material in tape form intended for use as a filler, material for splicing power and control cables as shown in section 4 of 9000-S6202-73980.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

MILITARY
MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts: Preparation for Delivery of.

DRAWING

NAVAL SHIP ENGINEERING CENTER
9000-S6202-73980 - Systems; Standards, Electric Plant Installations, Standard Methods.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material description. The filler shall be supplied in a roll as a tape, and have a suitable separator to facilitate application and insure uniformity of application. It shall be compounded of 100 percent synthetic materials designed to give the maximum properties as to performance as specified in 3.3. The filler shall be free of lumps and voids. A liner shall be employed which shall be easily removable during application.

3.2 Filler tape. Filler tape shall be furnished in the size shown in table I.

Table I - Sizes.

Thickness (maximum)	Width ± 1/16 inch	Length (feet)
Inch	Inches	Feet
0.125	1-1/2	5

3.3 Performance.

3.3.1 Application. The material shall have application characteristics describable as putty-like. This means that it shall be easily moldable with reasonable finger pressure or applied winding pressure. It shall have maximum stretch characteristics containing a minimum of recovery tensions, yet be able to maintain

FSC 5970

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a continuity of material that will not crumble, flake thin apart into stringiness, or otherwise react to cause an unintended break in its normal application.

3.3.2 Reaction. The material shall be so compounded as to be stable and long-aging in its stored form, and when applied onto its ownself and from its own applied pressure shall form into a stable homogeneous mass in not more than 6 hours. The reaction must be of its own composition and shall not be dependent wholly or partially to any degree on any other materials, chemicals, extraneous processes, or operations being applied on or within the material.

3.3.3 Properties. The material shall have the performance characteristics specified in table II.

Table II - Performance characteristics.

Dielectric strength	200 volts per mil.
Insulation resistance (96 percent relative humidity)	100,000 megohms
Electrolytic corrosion factor	1.00
Tensile	8 pounds per inch width
Elongation	1,000 percent

3.3.4 Flameproofness. The material shall be flameproof in its stored form, in its application, and in use. Determination of flameproofness will be based on test specified in 4.3.2.

3.3.5 Other performance. The material shall perform satisfactorily when utilized with other materials, as specified in section 4 of 9000-S6202-73980.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. For purposes of sampling and inspection, a lot shall consist of not less than 500 nor more than 1,000 rolls manufactured under substantially identical conditions.

4.2.2 Sampling for quality conformance inspection. From each lot of tape, sample rolls shall be selected at random as shown in table III.

Table III - Sampling for quality conformance inspection.

Lot size number of rolls	Sample size, rolls
200 and under	1
201 to 500	2
501 to 1000	3

4.3 Quality conformance inspection. Samples shall be subjected to the examination and test specified in 4.3.1 and 4.3.2 to determine conformance with this specification. Any sample not conforming will be considered unacceptable material.

4.3.1 Examination. Each sample roll selected in accordance with 4.2.2 shall be unrolled and measured and applied to a piece of electrical cable as for a repair.

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4.3.2 Flame test.

4.3.2.1 Procedure. A piece of the insulation tape 1 inch square shall be held over a gas flame (Bunsen burner) until there is evidence that the material itself is burning of its own accord. It will then be removed from the flame and observation will be made as to its ability to support its own combustion. It shall be considered undesirable to have combustion continue in excess of 5 seconds from time of removal from the burner. Material shall further display the property of remaining in its original form as an unbroken mass although charred on the surface.

4.3.3 Tensile strength and elongation test.

4.3.3.1 Procedure. Remove a 3 inch to 5 inch piece of filler tape from the liner without elongating it. Wrap two layers of pressure-sensitive crepe paper tape around one end of the filler tape perpendicular to its length. Wrap another two layers of pressure-sensitive crepe paper tape 1 inch away from the first. Clamp the paper tape in flat jaws of the tensile machine using a 1 inch jaw separation. Elongate the filler tape at a rate of 12 ± 1/2 inches/minute. Record the maximum tensile value which occurs (this occurs before the filler tape is elongated 100 percent), not the tensile strength at break.

Note 1: If 1-1/2 inch wide jaws are not available, slit the filler tape down to 1 inch, before removing the liner, with a sharp razor blade. The ring of crepe paper tape is vital in order to prevent the filler tape from slipping out of the jaws. If difficulty is encountered during the test, periodically tighten the jaws to prevent slippage.

Note 2: To prevent slippage, jaws of tensile machine may be lined with rubber.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements.)

5.1 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

5.2 Preservation and packaging, packing, and marking.

5.2.1 The electrical insulating tape shall be preserved and packaged, packed, and marked in accordance with MIL-E-17555. The tape shall be wound in rolls containing the length specified in 3.2.

5.2.2 Additional marking. Each roll shall be durably and legibly marked with the following, in such a manner that the markings will not become damaged when the packages are opened:

(a) Date of manufacture.[1/]
(b) Each packaged roll and shipping container shall be marked with the following CAUTION:

"Store in original boxes and in a cool, dark location. Do not store in close proximity to steam pipes, radiators or other sources of heat."

1/ Date of manufacture is defined as the date on which the actual manufacturing process was completed.

6. NOTES

6.1 Intended use. The insulation tape covered by this specification is specified for use as filler material in shipboard electric power and control cable splices.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Where a manufacturer's insulating tape of the same manufacture, material, thickness, and width has passed all tests on a previous submission, only the examination of 4.3.1 will be required.

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6.3 International standardization agreement. Certain provisions of this specification are the subject of international standardization agreement ABC-NAVY-STD-17C. When amendment, revision, or cancellation of this specification is proposed which will effect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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